

VerbaTask

Autonomous Voice-First WhatsApp ERP & Conversational CRM for
Micro-Merchants

Project Proposal & System Design Report

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Prepared for:

Bano Qabil x Alibaba Cloud AI Hackathon 2026

Track: Small & Mid-Sized Business (SMB) Retail Technology

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1. Executive Summary

VerbaTask is a zero-screen, zero-typing, voice-first conversational commerce engine built directly on WhatsApp for small and mid-sized retail businesses in Pakistan, neighborhood grocery (kiriyana) stores, apparel boutiques, pharmacies and general electronics or mobile-accessory shops. Rather than asking a merchant to learn a new application, VerbaTask meets the merchant inside the messaging app they already use every day and lets them run their entire sales, stock and payment-reconciliation workflow through natural spoken or typed conversation in Urdu, Roman Urdu or English.

The system's central innovation is a true voice-in, voice-out conversational loop: a merchant sends a voice note describing a sale or an automation and VerbaTask replies with a spoken voice note in the same language, alongside a lightweight text receipt for record-keeping. Combined with an automatic spell-correction and cross-lingual item-matching engine, a comprehensive registry of Pakistani payment channels and a merchant-configurable settings module, VerbaTask removes every barrier, literacy, language, hardware cost and technical setup, that has historically kept this market segment locked out of digital business tools.

This report documents the complete system design, architecture, feature set, database schema and security posture of the working VerbaTask prototype as submitted to the Bano Qabil x Alibaba Cloud AI Hackathon 2026.

2. Problem Statement & Market Analysis

2.1 The Kiriyana / Micro-Retail Reality in Pakistan

Millions of small and mid-sized merchants across Pakistan, kiriyana store owners, apparel and general retail shopkeepers, pharmacists and mobile/electronics vendors, manage day-to-day operations manually, tracking sales and customer credit (khata) on paper registers and reconciling digital payment confirmations by eye against a mental record of open orders. This segment sits entirely outside the reach of traditional ERP and point-of-sale platforms such as SAP, Odoo or desktop POS systems, for reasons that are structural rather than incidental:

- High barrier to entry: traditional systems assume dedicated hardware (POS terminals, laptops) and stable broadband, both of which are costly or unreliable for a single-owner retail shop.
- Language and literacy barriers: many shopkeepers are far more comfortable speaking Urdu than reading or typing English and a meaningful share are not comfortable with sustained reading or typing in any script.
- Existing WhatsApp bots are inadequate: current-generation bots may accept a voice note as input, but reply with dense blocks of English text, for a merchant serving a customer at the counter, reading a wall of text on a small screen defeats the purpose of speaking in the first place.
- Spelling and cross-lingual fragmentation: semi-literate merchants phonetically misspell items in English or Roman Urdu (e.g. "rise" or "riece" for Rice, "sugr" or "cheni" for Sugar, "coking oel" for Cooking Oil, "aata" for Flour). Stock is frequently catalogued in English while orders are spoken in colloquial Urdu ("2 kilo chawal" or the equivalent in Urdu script). Systems relying on exact-match string search fail constantly with "item not found" errors against this exact population.
- Fragmented local payments: Pakistan's retail payment landscape spans cash, mobile wallets and a wide range of commercial and Islamic banks (Easypaisa, JazzCash, SadaPay, NayaPay, Raast P2P, Meezan Bank, HBL, Bank Alfalah and others). A generic, one-size-fits-all payment list forces merchants to sift through channels they do not actually accept.

2.2 Why This Is a Genuine Gap, Not an Incremental Improvement

The failure mode here is not "the software is inconvenient", it is that the entire category of tool assumes a literate, English-comfortable, desktop-oriented user, which excludes a very large share of Pakistan's actual retail workforce. VerbaTask is scoped specifically to close this gap for single-location small and mid-sized businesses; it deliberately excludes enterprise concerns such as multi-branch synchronization, multi-warehouse logistics or large cash-and-carry chain operations, which are already served, however imperfectly, by existing enterprise tooling.

3. Proposed Solution & Value Proposition

3.1 The "Voice-In, Voice-Out" Paradigm Shift

VerbaTask's core design bet is that automation for this market should be spoken, not configured. A merchant interacts with VerbaTask exactly as they would with a human shop assistant standing beside them: they describe what happened, in whichever language and however they naturally phrase it and VerbaTask does the rest, resolving what item was meant, deducting stock, recording the payment channel and replying back in a spoken voice note the merchant can understand without ever looking at a screen.

This reframes VerbaTask from a "chatbot with menus" into what is functionally a zero-training-cost ERP: no onboarding session, no manual, no literacy requirement and no English requirement, because the entire interface is conversation in the merchant's own language and voice.

3.2 Value Proposition Summary

Stakeholder	Value delivered
Small/mid retail merchant	Runs sales, stock and payment tracking entirely by talking to WhatsApp, no new app, no typing, no English required, works on the cheapest available Android phone.
Merchant's customers	Faster, less error-prone checkout; payment channel options match exactly what the specific shop accepts.
Hackathon evaluators / judges	A genuinely novel, technically deep application of Alibaba Cloud's Qwen-2.5 language model combined with a production-grade voice pipeline, targeted at a large, underserved real-world population.

4. System Architecture & Data Flow

4.1 High-Level Architecture

The system is composed of four cooperating layers:

- Messaging Gateway, Meta WhatsApp Cloud API (Graph API v20.0), handling inbound/outbound messages, interactive buttons and lists and audio media, secured with SHA-256 HMAC webhook signature verification.

- Backend Core, Node.js (ES Modules) on Express.js orchestrating the conversational state machine, the voice pipeline, cross-lingual item resolution, payment-method registry and CRM logic.
- AI Services, Groq-hosted Whisper Large-v3 for speech-to-text, Alibaba Qwen-2.5 (via DashScope / Groq) for natural-language intent extraction and semantic item resolution and a multi-tier text-to-speech pipeline for the spoken reply.
- Data & Real-Time Layer, MongoDB (via Mongoose) as the system of record, with a Socket.IO WebSocket channel pushing live updates to the web dashboard as WhatsApp activity occurs.

4.2 System Architecture Breakdown

Step	Component / Service	Data / Action	Next Destination
1	Merchant (WhatsApp)	Sends Voice Note / Text	Meta WhatsApp Cloud API
2	Meta WhatsApp Cloud API	Forward via HMAC-signed webhook	Backend Core: Express.js
3a	Backend Core (Express.js)	If Audio: Forward audio payload	Groq Whisper Large-v3 STT
		If Text / Structured Data	Qwen-2.5 NLP + Cross-Lingual Item Resolver
4	Groq Whisper Large-v3 STT	Send Transcript	Qwen-2.5 NLP + Cross-Lingual Item Resolver
5	Qwen-2.5 NLP + Item Resolver	Processed Text / Entities	CRM / Order / Workflow Engine + MongoDB
6	CRM / Workflow Engine	Real-time Update Push	Socket.IO $\rightarrow$ Web Dashboard
		Response Text	Multi-tier TTS (Edge / Gemini / ElevenLabs)
7	Multi-tier TTS	Audio Stream / PCM	lamejs PCM $\rightarrow$ MP3 Encoder
8	lamejs Encoder	audio/mpeg Payload	Meta WhatsApp Cloud API
9	Meta WhatsApp Cloud API	Send Voice / Text Response	Merchant

4.3 The Spoken Voice Pipeline

The voice-in, voice-out loop is the system’s signature technical achievement, chaining together speech recognition, language understanding, business-logic execution and speech synthesis into a single sub-few-second round trip:

Stage	Component	Function
1. Capture	WhatsApp Cloud API	Merchant’s voice note arrives as an audio media object on the inbound webhook.
2. Transcribe	Groq-hosted Whisper Large-v3	Audio is transcribed with domain-tuned prompts covering kiryana-retail vocabulary (chawal, aata, daal, chini, ghee, sadapay, nayapay, raast, etc.) to improve accuracy on local terms.
3. Understand	Qwen-2.5	The transcript is parsed into a structured command (log a sale, create an automation, answer a question), with few-shot prompting for intent extraction and business-entity recognition.
4. Resolve	Cross-lingual item matcher	The named item is matched against the merchant’s catalog via the bilingual canonical dictionary, phonetic simplifier and LLM fallback described in section 4.3.
5. Execute	CRM / Order engine	Stock is deducted, the order is recorded and any triggered workflows or approval requirements are evaluated.
6. Synthesize	Multi-tier TTS pipeline	A spoken confirmation is generated in the merchant’s language via the highest-available tier (Gemini → ElevenLabs → Edge TTS → Google TTS fallback).
7. Encode & deliver	lamejs (pure-JS MP3 encoder)	Raw 16-bit 24kHz PCM audio is encoded to a valid audio/mpeg stream meeting Meta WhatsApp Cloud API’s media constraints, then uploaded and sent back as a WhatsApp voice message, alongside a short text receipt.

4.4 Cross-Lingual Semantic Matching Algorithm

Item resolution runs as a three-stage cascade ordered from cheapest/fastest to most expensive, so that the common case resolves instantly and only genuinely ambiguous cases incur an LLM call:

- Stage 1, Canonical bilingual dictionary: a maintained lookup table linking English, Roman Urdu and Urdu-script forms of common retail items (e.g. Rice ↔ Chawal ↔ چاول), resolved with 0ms local latency.

- Stage 2, Phonetic simplifier: common misspellings and phonetic variants produced by semi-literate typing or transliteration ("rise"/"riece" → Rice, "sugr"/"cheni" → Sugar) are normalized against the merchant's actual catalog entries.
- Stage 3, Qwen semantic fallback: for rare local brand names or genuinely ambiguous references that the first two stages cannot resolve, an inline Qwen call (resolveItemName) attempts semantic resolution against the merchant's catalog rather than surfacing a hard "item not found" failure.

This cascade is what allows a system trained primarily on structured data to remain usable for a population that does not produce clean, structured input.

5. Detailed Feature Breakdown

5.1 Voice-In, Voice-Out Spoken Urdu/English Interaction

Every conversational turn supports both a typed and a spoken path, converging on an identical internal command representation and every reply can be delivered as a spoken voice note in the merchant's selected language in addition to a text receipt for record-keeping.

5.2 Automatic Spell-Correction & Cross-Lingual Item Resolution

Described in detail in section 4.3, implemented in the item-matching and order-service modules, this is the layer that makes the system tolerant of real-world, non-standardized input rather than requiring a merchant to type items exactly as catalogued.

5.3 Complete Pakistani Payment Platform & Banking System

A registry of over twenty Pakistani payment channels, including SadaPay, NayaPay, Raast, Meezan Bank, HBL, UBL and Bank Alfalah, with alias normalization (so a merchant or customer can refer to a channel by any common name) and Urdu speech formatting for use in spoken confirmations.

5.4 Merchant Profiling & Settings Module

Merchants can rename their store, set core business details and, critically, choose exactly which payment channels their specific shop accepts, via the merchant profile and payment-methods endpoints. This directly drives which options appear in the WhatsApp guided ordering flow, so a merchant who only accepts cash and EasyPaisa is never shown a JazzCash or bank-transfer button.

5.5 WhatsApp Guided Order Flow with Dynamic Interactive Buttons

For merchants who prefer tapping over speaking, a fully guided flow presents an interactive item list drawn live from the merchant's own stock, followed by quantity entry, followed by a payment-method button set that is dynamically generated from that merchant's configured payment methods rather than a fixed, generic list.

5.6 Stock Management, High-Value Order Approvals and Autonomous Workflows

Every sale, however logged, deducts against a single shared inventory record. Orders that exceed a configurable value or risk threshold are routed into a human-in-the-loop approval flow rather than executing automatically and merchants can define plain-language automations (e.g. stock-threshold alerts) that the system evaluates and fires without further manual input.

6. Database Schema & Data Models

Model	Key fields	Purpose
Merchant	whatsappNumber, email, businessName, language, paymentMethods, onboardingComplete	The core account record linking a WhatsApp number to a web-accessible profile and its configuration.
InventoryItem	merchantId, name, quantity, price, unit	A catalogued product or stocked item belonging to one merchant.
Order	merchantId, items[], total, paymentMethod, status, source	A completed or pending sale, tagged with which input path (guided, voice, typed) produced it.
ConversationState	whatsappNumber, flow, step, data	Tracks a merchant's progress through a multi-turn flow (onboarding, guided ordering) between WhatsApp messages.
Workflow	merchantId, trigger, condition, action, active	A merchant-defined automation, message, schedule or threshold triggered.
Approval	merchantId, type, refId, status	A pending human-in-the-loop decision on a high-value or high-risk action.

7. Security, Validation & Error Handling

- Webhook signature verification: every inbound WhatsApp webhook event is validated against a SHA-256 HMAC signature computed from the app secret over the raw request body, rejecting any request that is not genuinely from Meta.
- Idempotent message handling: inbound message IDs are tracked to silently drop duplicate deliveries, which Meta's infrastructure can and does send on slow or non-200 responses.
- Graceful AI-service fallbacks: the text-to-speech pipeline is tiered (Gemini → ElevenLabs → Edge TTS → Google TTS) so that a single provider outage degrades quality rather than breaking the reply entirely; the item-resolution cascade similarly degrades from instant dictionary lookup to LLM fallback rather than failing outright.
- Input validation on every REST endpoint, with a consistent success/error response envelope so client-side error handling is uniform across the API surface.
- Automated test coverage: 39 automated tests across 6 suites validate core business logic (order creation, stock deduction, payment-method handling and related flows) ahead of every deployment.

8. Impact, Usability & Future Roadmap

8.1 Impact

By removing literacy, language and hardware barriers simultaneously, VerbaTask targets a segment of Pakistan's retail economy that is large, economically significant and structurally excluded from existing digital tooling, not due to lack of willingness to adopt technology, but because existing technology was never designed with this user in mind.

8.2 Usability

The entire system requires no app installation and no training: a merchant's first interaction with VerbaTask is identical to sending a WhatsApp message to a person, which is already a familiar action for effectively the entire target population.

8.3 Future Roadmap

- Customer-facing digital receipts sent directly to the buyer's WhatsApp number.
- Proactive daily/weekly business-performance voice digests.
- Expansion into additional regional languages (Punjabi, Pashto, Sindhi).
- Supplier-facing automated restock messaging when stock crosses a merchant-defined threshold.

These items are intentionally scoped as roadmap rather than current functionality and are described here to indicate architectural direction, not present capability.